Jackson Small

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Education

University of Central Florida Expected May 2028
M.S. in Statistics and Data Science | Graduate Teaching Assistant *Orlando, FL*

Award: Graduate Teaching Assistantship ($15,000 stipend) plus Graduate Dean's Fellowship ($5,000 competitive merit award), $20,000 total with full tuition remission and health insurance

University of Central Florida, Burnett Honors College May 2026
B.S. in Data Science, Minor in Statistics, GPA: **3.83** *Orlando, FL*

Coursework: Statistical Learning, Applied Time Series, Stochastic Processes, Mathematical Modeling, Statistical Theory I & II, Big Data Analytics, Matrix & Linear Algebra

Technical Skills

ML/AI & Statistics: PyTorch, Scikit-learn, CNNs, QLoRA Fine-Tuning, RAG/LLMs, Statistical Modeling, Time Series Analysis, SMOTE, Cross-Validation, A/B Testing
Programming & Tools: Python, R, SQL, C#, Julia, Bash, Jupyter, Git/GitHub, Docker, Streamlit, Power BI, Azure OpenAI, SLURM/HPC
Methodologies: ML Pipelines, Feature Engineering, Signal Processing, Experimental Design, Model Evaluation, Anomaly Detection, HIPAA Compliance

Experience

AI Associate Developer Oct 2025 – Aug 2026
Insurity, Inc. *Remote*

- Directed AI coding agents (Claude, Replit) as sole engineer across six production workstreams on a .NET/SQL enterprise claims platform, delivering 37,600+ net lines of C#, SQL, and Python across 11 months with full architectural ownership, code review, and test coverage of all generated output.
- Engineered 5 Python automation tools to migrate 175 enterprise reports to Power BI across 6 business lines in 3 weeks against a 3-4 month projection, parsing 264,000+ lines of XML to apply 1,225 modifications across 187 database connections with a 100% verification pass rate, zero production incidents, and $700,000 in cost avoidance.
- Architected a dual-layer HIPAA audit system running on 23 PHI tables (9,594 lines of C# and SQL), using dynamic SQL triggers with runtime column discovery that cut per-record storage 85-96% (1,745 to 186 bytes) and eliminated 280+ lines of static code per table, backed by a 5-step username resolution chain and 3-tier role-based access control.
- Built a Roslyn static analysis tool that parsed a legacy C# codebase into a type-resolved inventory of every data-layer call site, then drove a Python codemod that automated the refactor across all matched sites, replacing manual auditing with a programmatically verified transform.
- Delivered an enterprise document-storage integration in 22 days against a 3-6 month industry standard, architecting a provider-pattern abstraction with three interchangeable storage backends, OAuth2 client-credentials lifecycle management, and DualWrite zero-downtime migration, validated by 234 automated tests across 16 REST API route families.

Research Assistant Intern May 2024 – Aug 2024
Design Interactive, Inc. *Orlando, FL*

- Designed automated data validation pipelines using Python and SQL for large-scale human systems research datasets, ensuring statistical integrity across experimental conditions.
- Led quality testing of VR/AR simulations in Unity, deploying builds to hardware and validating eye-tracking software to capture participant gaze patterns for behavioral research.

Research

Undergraduate Researcher Mar 2026 – Aug 2026
UCF AIM Lab (AI & Imaging in Medicine) *Orlando, FL*

- Formalized a new computer-vision task, video anomaly anticipation (predicting anomalies in the unobserved future, before they are visible on screen), distinct from the standard detection setting, and established the first anticipation baselines on the UCF-Crime and XD-Violence surveillance benchmarks.

- Built a method that reads a frozen 5B-to-14B-parameter video generative model (Wan 2.2) via flow-matching ODE inversion, training only an 11M-parameter head on the model's intermediate latents (under 0.2% of the backbone) with no fine-tuning, and recovered a statistically significant early-warning signal (within-window AUC 0.591 vs 0.411 control, paired +0.180, $p < 10^{-4}$ across all 10 random seeds).
- Benchmarked 5+ frozen vision-language foundation models zero-shot (InternVL3-78B, LLaVA-OneVision, VideoLLaMA3, Qwen2.5-VL) on 14,000+ standardized video windows on the UCF Newton HPC cluster (H100 80GB, SLURM), designed a confound-immune evaluation protocol, and authored a 12-page conference-style paper draft handed off to the lab.

Projects

Honors Undergraduate Thesis (Published) | *Python, PyTorch, Signal Processing* Jun 2025 – Jan 2026

- **First Data Science undergraduate to publish in the UCF STARS Honors Undergraduate Thesis repository.** Authored *Theoretical Analysis of CNNs for Automatic Seizure Detection in EEG Signals*, advised by Dr. Chudamani Poudyal (SDMSS, UCF).
- Developed a 1D CNN with a Butterworth noise-filtering pipeline and SMOTE class rebalancing, achieving 97% accuracy and 0.99 AUC on high-frequency EEG time-series data.
- Established theoretical Lipschitz stability bounds ($L = 24.72$) to formally prove model robustness against real-world EEG signal noise.

DataSci-Coder: Fine-Tuned Code Generation Model | *Python, PyTorch, QLoRA, Hugging Face* Spring 2026

- Fine-tuned Qwen2.5-Coder-14B using QLoRA 4-bit quantization on 10,795 curated data science examples spanning statistics, ML, and deep learning, training on a single L40S GPU in under 2 hours.
- Built a multi-source data pipeline merging newsletters, coursework, and public datasets with automated deduplication, category balancing, and code-quality filtering.
- Achieved 93.3% instruction compliance against 91.4% for the base model, with 10/10 code-only output compliance versus 6/10 base and a 100% code ratio versus 87.9% base.
- Designed a dual evaluation framework combining a 12-task hard eval with a 10-test constraint eval measuring instruction-following across data cleaning, ML pipelines, statistics, deep learning, and visualization.

AI Driver Risk Scoring Platform | *Python, Scikit-learn, Streamlit, Pandas* Fall 2025

- Engineered 47 behavioral risk features from simulated telematics data to train predictive ML models achieving ROC-AUC 0.98 on driver risk classification.
- Deployed a real-time Streamlit dashboard with interactive Plotly maps, SHAP feature importance attribution, and gamified driver scoring for predictive visualization.

E-Commerce Shopper Behavior Analysis | *Python, K-Means, Random Forest, SMOTE* Oct 2025 – Dec 2025

- Analyzed 12,330 e-commerce sessions to predict purchase intent, applying K-Means clustering to discover 5 distinct user segments and SMOTE to correct severe class imbalance.
- Applied ANOVA and Chi-Square testing for feature selection, training a Random Forest classifier that achieved 0.93 AUC and 91% F1-score to extract actionable marketing insights.

Leadership & Certifications

Leadership: Eagle Scout (BSA) | Epilepsy Foundation Ambassador | Burnett Honors Scholar | CRU at UCF, Youth Group Leader & Worship Musician, Fall 2022 to Spring 2026
Conferences: AMS Spring Southeastern Sectional Meeting, Florida State University, March 2024
Certifications: Microsoft Azure AI Essentials | Anaconda Python for Data Science